EDUCATION

Program	Institution	CGPA	Year
B.E Computer Science Engineering (Ongoing)	Rajalakshmi Engineering College	8.3 Till 2nd semester	2028
B.S Data Science and Applications (Ongoing)	IIT Madras (Online)	6.0 upto 4 courses	Self Paced
X|| (CBSE)	Everwin Vidhyashram	84%	2024

TECHNICAL INVOLVEMENTS AND INITIATIVES

- Built a custom Arch Linux operating system with themed interface, voice unlock, and local AI.
- Active SEDS space club member contributing to microgravity projects, space technology content, and technical outreach
- Developed sustainability-focused software solutions for carbon tracking, food waste reduction, and automated attendance systems.

RELEVANT COURSE WORK

Data Structures and Algorithm	Application Development	Automation & Workflow Engineering
Database Management System	API Development	Web Technologies
Operating System	Python Programming	Object Oriented Programming

PROFESSIONAL EXPERIENCE

Independent Software Engineer	Tools: Python, FastAPI, JavaScript, MongoDB, Git, Linux ● Built full-stack applications for real-world use cases. ● Developed Linux-level automation tools and AI-powered workflows. ●Integrated local AI models into production-style applications..
Linux & AI Systems Developer	Tools: Arch Linux, Bash, XFCE, Python, Ollama, Mistral ● Built a custom Linux distribution with themed UI and system branding. ● Implemented voice-based authentication and AI-controlled system actions. ● Implemented voice-based authentication and AI-controlled system actions.
Backend & Automation Engineer	Tools: FastAPI, MongoDB, n8n, JavaScript, Docker ● Built REST APIs and database-driven backend services. ● Created QR-based attendance automation with verification and logging. ● Designed workflow bots for task automation and user interactions.
Hackathon	● Built a **proxy-resistant QR attendance system** for classroom and event verification. ● Selected **College-Level Round 2** in **Smart India Hackathon (SIH)** for attendance automation solution.

PROJECT WORK

Office Complaint Management System (AI-Assisted CRM)	Tools: Python, Pytorch, Flask, SQLite, HTML, CSS Developed a secure office complaint and ticketing platform with AI-based routing, priority handling, and role-based access control.
OD (On-Duty) Management System	Tools: Python, Flask, SQLite, HTML, CSS, n8n Built an automated system to handle student OD requests, approvals, and records using backend APIs and workflow automation.
TourMate - Smart Travel Comapnion app	Tools: Flutter, Google Maps API, REST APIs Designed a travel companion app that provides recommended places, nearby stays, fuel stations, hotels, itineraries, and expense tracking.
Proxy-Resistant QR Attendance System	Tools: React Native, SQLite, Python, ComputerVision(OpenCV) Created a secure QR-based attendance system that prevents proxy marking and ensures real-time verification and logging.
Love OS – Custom Linux Distribution	Tools: Arch Linux, Bash, XFCE, Python, Ollama Built a customized Arch Linux operating system featuring themed UI, voice-based login, and integrated local AI capabilities.

POSITION OF RESPONSIBILITIES

App Team - IEEE CS	Worked on building and improving application features for IEEE technical projects.
Content Team - SEDS	Created space-science and technology content for club activities and outreach.
Research Team - Enactus REC	Supported research and data collection for social-impact and sustainability projects.

PROFESSIONAL SUMMARY

Profile	Computer Science Engineering student focused on backend systems, Linux engineering, and applied AI.

GitHub:https://github.com/venkatengineer
LinkedIn: https://www.linkedin.com/in/venkat-badhrinarayanan-747148264/